

A Multi-Agent Energy Trading Competition

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Abstract

The energy sector will undergo fundamental changes over the next ten years. Prices for fossil energy resources are continuously increasing, there is an urgent need to reduce CO₂ emissions, and the United States and European Union are strongly motivated to become more independent from foreign energy imports. These factors will lead to installation of large numbers of distributed renewable energy generators, which are often intermittent in nature. This trend conflicts with the current power grid control infrastructure and strategies, where a few centralized control centers manage a limited number of large power plants such that their output meets the energy demands in real time. As the proportion of distributed and intermittent generation capacity increases, this task becomes much harder, especially as the local and regional distribution grids where renewable energy generators are usually installed are currently virtually unmanaged, lack real time metering and are not built to cope with power flow inversions (yet).

All this is about to change, and so the control strategies must be adapted accordingly. While the hierarchical command-and-control approach served well in a world with a few large scale generation facilities and many small consumers, a more flexible, decentralized, and self-organizing control infrastructure will have to be developed that can be actively managed to balance both the large grid as a whole, as well as the many lower voltage sub-grids.

We propose a competitive simulation testbed to stimulate research and development of electronic agents that help manage these tasks. Participants in the competition will develop intelligent agents that are responsible to level energy supply from generators with energy demand from consumers. The competition is designed to closely model reality by bootstrapping the simulation environment with real historic load, generation, and weather data. The simulation environment will provide a low-risk platform that combines simulated markets and real-world data to develop solutions that can be applied to help building the self-organizing intelligent energy grid of the future.

1 Introduction

By 2020, about 35% of the overall energy demand in the European Union, EU, will be generated through distributed and intermittent “green energy” resources [11]. This presents a severe challenge to the existing energy infrastructure, which was designed to distribute power from a few large generating plants. It is thus very important to adopt a “smart” management approach that will facilitate effective integration of these resources into the existing energy generation and distribution infrastructure. Currently, much of the distributed generation capacity installed is virtually unmanaged, and cannot be supervised from centralized grid control and power dispatch systems.

With a small share of renewables in the overall generation capacity mix, this integration has not been a problem. But this strategy has almost reached its limits. Grid balancing capacities within the European Union for the Coordination of Transmission of Electricity (UCTE) high voltage grid are designed to cope with short-term variations in generator output of at most 3000 MW. Planned investments and projects underway in renewable generator capacities are expected to increase overall generation volatility within the next five years to well beyond this level [26]. Furthermore, the number of renewable generators and their wide-spread distribution will strongly increase. In Germany, for example, a roll-out of 100 000 distributed Combined Heat and Power (CHP) power plants has begun, with a planned overall installed capacity of 2 GW¹. A purely centralized command-and-control approach to managing the grid and its generators has

¹This is roughly the capacity equivalent of two nuclear power plants, see http://www.lichtblick.de/h/idee_302.php

reached its limit, with the number of generators and the volatility of their output increasing even further over the next decade.

One approach to addressing this problem is to use the resource-allocation power of a market to find a near-optimal balance between producers and consumers of electrical power at lower levels of the grid hierarchy, and to make the participants in this market responsible for real-time balancing at the local level. So far, there is limited experience from pilot projects and field studies that could guide design and operation of such regional markets [14, 3]. All these projects rely on “intelligent devices” and automation technology to facilitate or even automate energy management at both consumer and generator sites. The automation solutions proposed within these projects are “unchallenged” in the sense that no other (competing) automation technology was deployed within these pilot projects, though competing technologies and solutions will be the default case in large-scale technology deployments on regional or national levels. The California energy market breakdown in 2000 [21, 5] is an example of the problems that can occur if potential strategic, competitive or collusive behavior of market participants is not sufficiently accounted for in the design of such markets. For these reasons, we propose and describe the design of a competitive simulation environment that uses a market-based management structure for a local energy grid, which would mirror reality fairly closely. This simulation environment would challenge research teams to create agents [47], or possibly agent-assisted decision support systems for human operators [44], that could operate effectively and profitably in direct competition with each other. Teams would also be challenged to exploit the structure of the market, and that structure would be adjusted periodically to defeat counterproductive strategic behaviors. The result would be a body of valuable research data that could guide energy policy, along with a much higher degree of confidence that such a mechanism could be safely introduced into operating energy systems.

We call this vision “TAC Energy” because it is an example of a Trading Agent Competition² applied to energy markets. In this context, TAC Energy will be designed to foster research in the operation of pricing and coordination strategies for the acquisition of energy from distributed resources, sale of energy

²see www.tradingagents.org

to distributed loads, and the real-time balancing of production resources with consumer energy loads. We assume that some portion of the load, including the charging and discharging of Plug-in Electric Vehicles (PEVs), could be controlled by voluntary or automated means, using prospective or real-time price signals.

The next section gives background on the structure and organization of current electrical energy production and distribution systems, and reviews previous work both in competitive simulations and in other approaches to addressing the control and resource allocation problems in future energy grids. Following sections explore the published literature that is relevant to this work, and describe major elements of the simulation design in some detail. We then outline the essential decision problems that must be solved by a successful competing agent, and conclude with a discussion of open research questions that can be addressed through a well-designed competitive simulation environment, many of which would be either impossible or highly risky to explore in the real world. While reading this paper, keep in mind that it is a proposal and not a completed, implemented design.

2 Background and Related Work

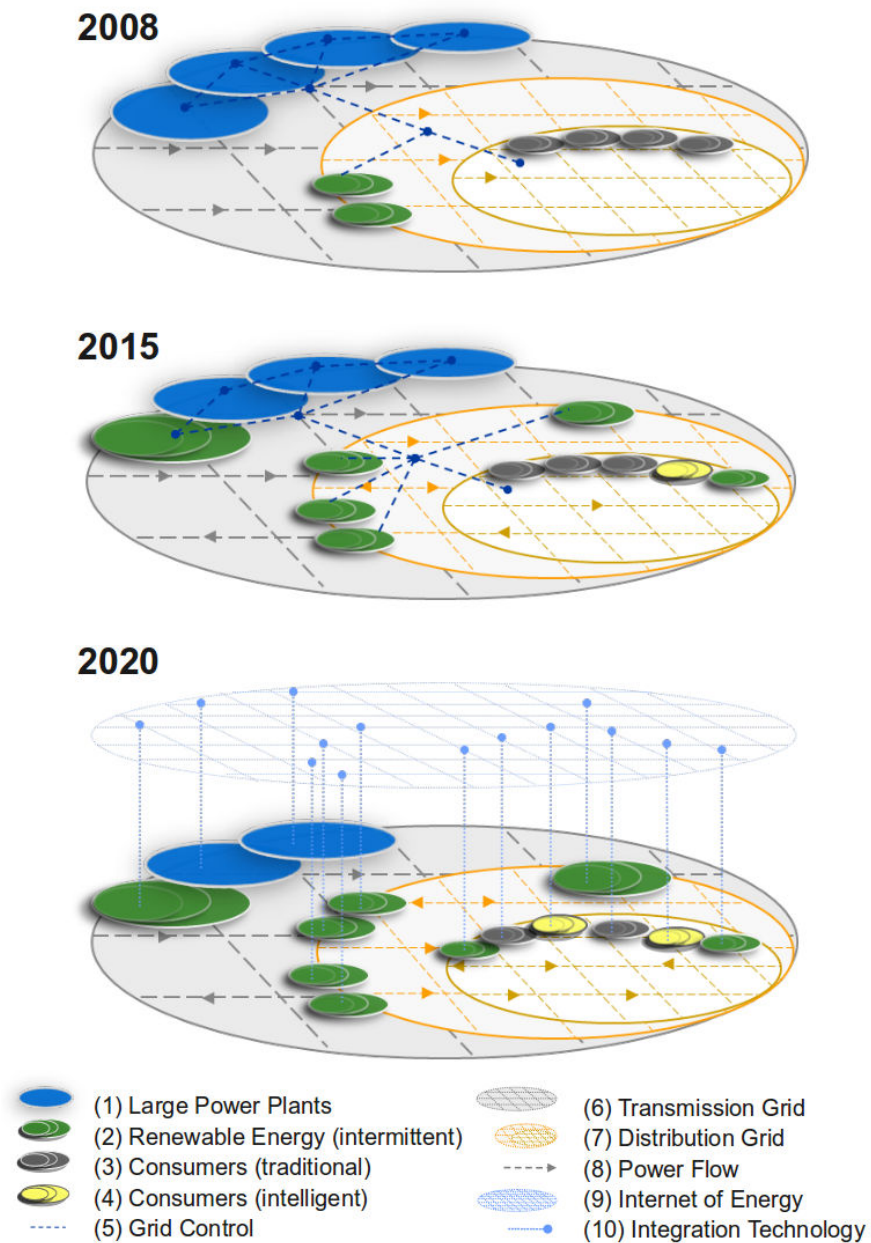
In this section we describe in more detail portions of the existing electrical energy distribution systems that are relevant to the discussion in this paper. We then discuss some of the principal industry and government initiatives that are intended to resolve the problems of integrating new energy sources into the grid, and we review related work in multi-agent modeling and competition testbeds.

2.1 Energy infrastructure

Traditionally, the electrical energy infrastructure is organized in a strict hierarchy: A few centralized control facilities manage relatively few large power plants and schedule their production according to forecast energy demands, which are usually based on synthetic load profiles, i.e. average historic consumption time series for different consumer groups. With these estimates at hand, generation capacities are dispatched. Figure 1 provides a simplified sketch of the current situation (2008). Power is transmitted from power

plants (1) to the consumers over a network of high, medium voltage transmission grids (6) into low voltage distribution grids (7) that provide the connection to end customers (3).

color, and the "integration technology" arcs are different in the



[Source: BDI Initiativ: ICT for Future Energy Markets (2008)]

Figure 1: Evolution of "Internet of Energy" from 2008 until 2020.

With an increasing share of distributed, renewable generation capacities installed in the medium and low voltage grids (2), generation volatility will increase and power flow inversions (9) will occur, which currently neither the technical grid infrastructure nor the control strategies can cope with effectively. In parallel, more and more consumers will be equipped with smart metering equipment and demand side management devices (DSM) that help them to monitor and actively manage their loads. Consequently, their consumption flexibility will increase and the predictability via synthetic load profiles becomes difficult, while at the same time they will become at least somewhat responsive to time-varying energy prices.

In order to be able to manage this complex and highly dynamic system, the existing energy grid will have to be transformed into a smart grid with secure real-time ICT integration and communication among all of its components (10 in Fig. 1). Based on this information and communication infrastructure, an “Internet of Energy” will evolve [4] that serves as a key enabler for new distributed and highly automated approaches to managing generators, consumer loads and the grid infrastructure.

On the business level, new market roles and business models will evolve. Similar to the evolution of smart business networks in supply chain management, virtual power plants (VPP) are likely to be established that bundle many small generators into one large virtual entity. A VPP would thus operate, dispatch, and sell its cumulative capacity to consumers. Besides the development of appropriate business models, major challenges on this level are the development of smart control strategies and (agent-based) automation technologies that facilitate the realization of these business models, services and products.

2.2 Initiatives

The U.S. National Institute of Standards and Technology (NIST) recently published the first draft of a “Smart Grid Interoperability Standards Roadmap” [9]. It defines a simplified domain model for a future smart grid with identified “Distribution”, “Market,” and “Customer” domains being in the core of the overall model. Furthermore a list of prioritized actions for the fast transformation of the current infrastructure into a smart grid is provided. Highest priority, according to NIST, are Demand Response and consumer energy

efficiency measures. In particular they state that *“Market information is currently not available to the customer domain. Without this information, customers cannot participate in the wholesale or retail markets. In order to include customers in the electricity marketplace, they need to understand when opportunities present themselves to bid into the marketplace and how much electricity is needed.”*

In October 2009 the EU Commission announced the Strategic Energy Technology Plan (SET Plan) [11] along with a draft technology roadmap [12]. One of the priority actions mentioned in this roadmap is the development of so called “smart cities” that efficiently and intelligently manage local energy production and consumption. In particular *“5-10 development and deployment programmes for smart grids in cities, in cooperation all relevant SET-Plan Initiatives, including priority access for local generation and renewable electricity, smart metering, storage, and demand response”* should be established within the next two years.

The BDI, a German power industry lobby group, published a technology roadmap that describes the transition from the current energy infrastructure into a so called “Internet of Energy” [4] on a timeline from 2009 to 2020. The document was written by a group of experts from various industries in cooperation with researchers from several different institutions. According to this roadmap, regional energy markets, virtual power plants based on micro CHP turbines, as well as DSM technologies will be mainstream by 2015. According to this document one of the key challenges will be the development of *“applications and services for coordinating the energy grid on the business level”* in a sense that the technical infrastructure will be in place but smart coordination and operation strategies are yet to be developed.

2.3 Multi-Agent Modeling

Electricity production and distribution systems are complex adaptive systems [28] that need to be managed in real time to balance the load of an electricity grid. Recently, electricity markets are undergoing or are about to undergo a transition from centrally regulated systems to decentralized markets [31]. These transitions are very risky since we do not have sufficient experience in setting up decentralized energy systems and predicting their effect on the economy. We have observed in recent history that failures in designing such

systems can cause major damage while deploying them in the real world. The California energy market [5], and the recent collapse of Enron, challenge the wisdom of deregulating the electricity industry, and have demonstrated that the success of competitive electricity markets crucially depends on market design, demand response, capacity reserves, financial risk management and reliability control along the electricity supply chain. Therefore, it is very important to thoroughly test system design proposals in a risk free simulated environment before deploying these ideas into the real world.

As electricity systems move more toward open, competitive markets, the need for complex modeling systems becomes more obvious. Although traditional optimization and simulation tools will continue to provide many useful insights into market operations, they are typically limited in their ability to adequately reflect the diversity of agents participating in these markets, each with unique business strategies, risk preferences, and decision processes. Rather than assuming we can predict the behavior of market participants, we suggest using agent-based tools in a laboratory setting. This framework will represent electricity markets with multiple agents, each with their own objectives, and decision rules. These agents will be developed by competing research teams, thereby avoiding the cognitive blindness that can limit the value of results arising from non-competitive experimental work in such simulation environments. Intelligent software agents [47, 17] offer many possibilities for automating, augmenting and coordinating business decision processes. These agents act on behalf of users, with some degree of independence or autonomy, employing some representation of the user's goals or desires. We focus on enhancing the adaptive learning component of such agents; our research is focused on trading agents that learn to operate effectively in competitive economic environments.

Agent-based modeling and simulation has emerged over the last few years as a dominant tool of the energy sector. For instance, the Electricity Market Complex Adaptive Systems Model (EMCAS) electric power simulation is an agent simulation that represents the behavior of an electric power system and the producers and consumers that work within it [31]. In [40], the authors have developed MAIS, an agent-based decision support system for analyzing and understanding dynamic price changes for the U.S. wholesale

electricity market before and during the California energy crisis. A nice review of agent-based simulation tools for energy markets can be found in [48].

The field of Agent-based Computational Economics (ACE) [42] is the computational study of economic processes modeled as dynamic systems of interacting agents. Here “agent” refers broadly to a bundle of data and behaviors that represents an entity in a computationally constructed world, as opposed to the individual trading agents that are the focus of our research. ACE is mostly focused on advising policy makers, and includes usually the following three main steps: (i) model the current real-world environment as closely as possibly with an agent-based simulation, (ii) change specific simulation parameters, such as market recession, and observe the effects, and (iii) based on the previous outcomes give recommendations to policy makers. A good example of ACE is given in [45], a large scale simulation of interrelated German energy markets. Policy recommendation based on ACE models have a qualitative rather than a quantitative nature. Focusing on the self-organizing features of economic systems does not necessarily imply a laissez faire approach to economic policy – an important question to ACE policy modelers is what institutional setting is best in supporting self-organizing features of the system.

2.4 Multi-Agent Competitions

Along with the vision and development of the “Internet of Energy,” we identified a lack of automation and intelligent strategies as problems in the future of electricity production and distribution. In this context, agents are a promising tool to help solve these issues. The proposed “TAC Energy” competition will provide a realistic, low-risk platform to effectively test and evaluate agent strategies and decision procedures. Over the last decade competitions are becoming increasingly prevalent in the research world. The current Trading Agent Competitions use a multi-year competition format to study trading in simultaneous online markets (TAC Classic) [46], operation of a three-tier supply chain (TAC SCM) [7], trading of search keywords for advertising purposes (TAC AA) [19], and the operation of online exchanges (TAC CAT) [30].

Each research methodology has strengths and limitations. A well designed competition has many ben-

efits [39]. Much has been learned through designing, developing, and researching TAC SCM over the last eight years. This experience will guide the design of TAC Energy. For instance, one of the awkward features of TAC SCM is the design of the start and end of the scenario; agents start with no inventory at the beginning, and inventory has no residual value at the end of the scenario. TAC Energy will not have these problems, since the aggregator agents need to balance the grid in real time with extremely limited opportunity to store energy. Another shortcoming of TAC SCM is that the simulation is completely driven using predetermined statistical distributions. These distributions and parameters were carefully designed to balance the game among the three main agent types (supplier, manufacturer, and customer), but nevertheless they have exploited in unrealistic ways by some research teams, and have generated much discussion among participants about the generalizability of results to real world scenarios. In the TAC Energy simulation we are in the fortunate position to have a large body of real world data from a four year pilot project, MEREGIO, covering 900 households in Germany [16].

TAC Energy has many more potential benefits as a research platform, such as the natural human desire to win. This will be a strong motivation to create a competitive agent by solving the challenges of the domain. Another advantage is that competitions force researchers to build complete, working systems by a specified date. Competitions such as TAC Energy can be strong research drivers for work in specific domains. For example, over the last seven years the annual TAC SCM tournament has pitted autonomous manufacturing agents against each other to determine which is the best performing agent. Each year, results from the previous competition have been published, such as [24, 2, 23], which rapidly raises the bar for research. Since all teams have to solve the same problems in the same domain, this becomes a strong driver for research, and different teams learn from each other. Teams are encouraged to contribute their agents to a shared repository, allowing researchers to run controlled experiments that demonstrate the effect of individual agent designs, such as described by [20, 37].

Market liberalization changes the landscape for corporate managers and public policy makers, who face difficulty in both predicting and understanding price changes in electricity markets. Price changes occur due

to many uncontrollable factors such as changes in weather conditions, demographic changes, and different trading strategies among traders. Therefore, we need tools to identify and predict market conditions in a dynamic fashion, such as the economic regimes method described in [23].

3 Competition Scenario

The main goals of the proposed simulation are (i) to provide a competitive testbed for the development and validation of a market structure for managing electrical power distribution in a local grid, (ii) to spur research and development on intelligent agents and decision support systems that help automate decision processes in such markets, and (iii) to ease knowledge transfer between research and application by providing a testing environment that closely resembles reality. The entities competing in this market will broker electrical power in a local energy grid that contains a mix of intermittent energy sources along with residential, commercial, and industrial demands.

The competition introduces the role of an “aggregator,” which will be represented by a trading agent. According to [9] “*aggregators combine smaller participants (as providers or customers or curtailment) to enable distributed resources to play in the larger markets.*”. In reality aggregators could be energy retailers or municipal utilities, or in some cases large utilities could also take on the role of aggregators.

Within the competition an aggregator sells tariff contracts to end customers (e.g. households, small and medium enterprises, owners of electric vehicles), which are attracted or deterred by the respective tariff conditions. Tariff conditions may include flat prices, time of use prices, peak prices, load caps for certain times of the day, contract runtime, etc. In addition to “classical” tariff contracts for energy consumption, an aggregator can also sell “energy production” tariffs to end customers. Under such a tariff agreement, a customer may be paid for operation of a decentralized energy generator such as a Combined Heat and Power (CHP) plant that feeds power into the grid.

Another type of special customer is Plug-in Electric Vehicles (PEVs). These customers receive special tariff contracts that have separate, time dependent prices for charging the vehicle (consuming energy) and

for feeding energy back into the grid (effectively producing energy). As compared to households, PEV customers are comparably large energy consumers during their charge cycle but might decide to discharge some of their stored energy at their own discretion if the power generation prices are sufficiently attractive. Aggregators may limit the maximum charge rate for PEVs (e.g. at most 16A), effectively throttling the speed of recharge, and they might also encourage PEVs to feed energy back into the grid by setting the generation price appropriately, but they cannot directly enforce charging or discharging.

On the tactical level (planning horizon: 1 week – 3 months) aggregators have to manage their portfolios of consumer, generator and PEV contracts. On the operational level (planning horizon: 1 day – 1 month) aggregators have to balance the fluctuating energy demands of their customers against the actual output of their contracted energy production capacity. Differences can be compensated through balancing power purchased or sold at an energy exchange.

The competition is designed to model most of these challenges as close to reality as possible while keeping computational and technical complexity manageable.

In particular we make the following *assumptions*:

1. Within the simulated region, grid constraints (line capacity limitations) are assumed to be non-existent, i.e. power flows within the region are unconstrained. Local distribution grids are typically overdimensioned with respect to their line capacities, thus this assumption is not a strong restriction but may have to be rethought in future once much more distributed generators and storage facilities are installed.
2. The point of common coupling (PCC) between the simulated distribution grid and the higher level transmission grid has a maximum capacity for power inflow and outflow. A specialised agent that serves as a “liquidity provider” on the regional energy spot market, and is able to arbitrage with the national energy spot market, has to obey these technical limits.
3. Power factor effects, i.e. phase shifts between voltage and current, are not taken into account. Modelling these effects would possibly influence the aggregators’ decision making on which consumers

and generators to add to their portfolios but is out of scope at this time.

4. Power distribution and transformation losses are ignored. In Germany these losses are estimated at 3%; for North America they are estimated at 5,5% [43]. These losses can be considered as being more or less constant within a distribution grid and identical for all grid participants. Thus the validity of the simulation results is not affected.
5. Two kinds of generators (energy production facilities) are distinguished. One kind (photovoltaic arrays, wind turbines) produce power when active, and are under control of their respective owners. The second kind (PEV batteries, some CHP units) is called “controllable” and may be switched on or off, or have its output adjusted remotely within its capacity range.
6. Technical load balancing (i.e. the real time operations of the local distribution grid) is accomplished outside the action domain of the competition participants (see Section 3.2.2) using a combination of controllable generators and spinning reserves.
7. The simulation will model time as a series of discrete “timeslots” rather than as continuous time.
8. The temporal distribution of energy consumption and generation *within* a timeslot is not taken into account. This means for example that balancing power demand for a timeslot is calculated as the difference of the sum of generation and the sum of consumption for that timeslot and not as the instantaneous difference between the two timeseries.

In order to expose the aggregator agents to tactical and operational decision making, the competition scenario proceeds through a series of alternating *contracting* and *operation* phases as depicted in Figure 2. Both phases are described in more detail in subsequent sections.

To enhance the realism of the competition scenario, we will drive the simulation with real historical data on generation, consumption, and weather information, along with a model of preferences of potential customers derived from customer surveys and pilot projects. One source of such data series is the German

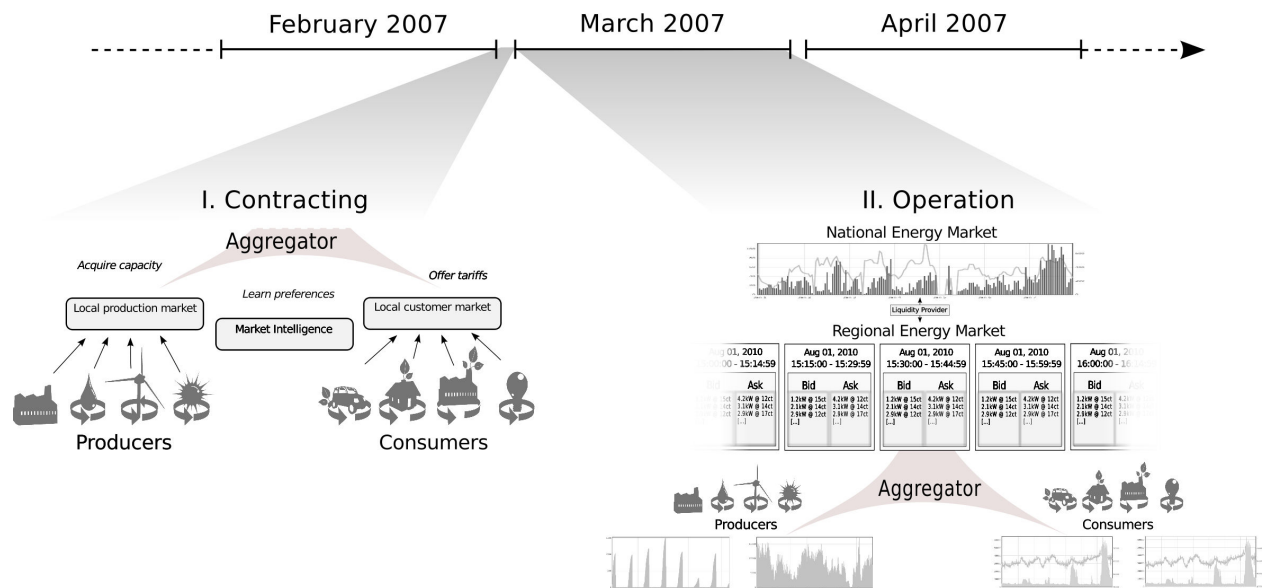


Figure 2: Simulation timeline

MeRegio project, a smart grid project that is implementing a combination of advanced grid control systems and innovative real-time pricing tariffs [16]. Figure 3 shows the area around Freiamt, which currently serves as a pilot region for MeRegio. Freiamt contains a range of different distributed renewable energy generation facilities in combination with households and small and medium enterprises (SMEs) that are currently equipped with demand side management devices allowing them to flexibly react to price signals from the distribution grid.

With historic consumption and generation data collected from a region like this, the proposed simulation environment exposes the aggregator agents to the challenge of managing virtual consumers and generators, which exhibit realistic energy consumption and generation patterns based on the history data.

3.1 Contracting phase

On the simulation timeline, a contracting phase represents a short period of time (perhaps 60 seconds). During this phase, aggregator agents try to acquire energy generation capacity from local producers and from the regional exchange, and sell energy tariffs to local customers as depicted in Figure 4.

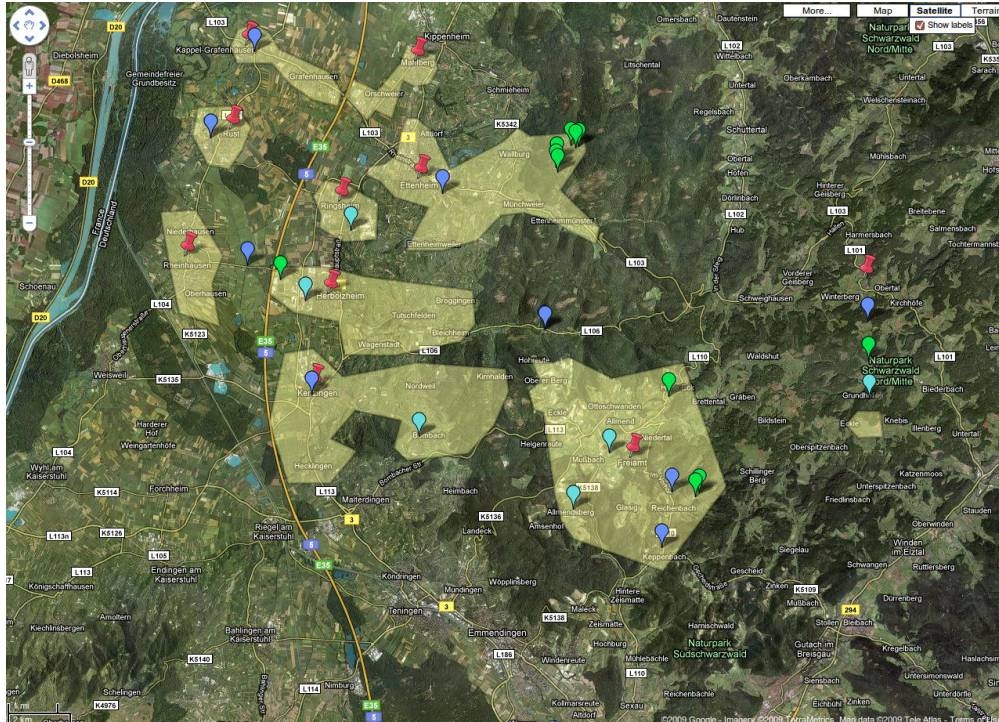


Figure 3: Freiamt, Germany: One of two pilot regions in the MeRegio project (Source: <http://maps.google.com>)

The development of a common semantic model and a common pricing model to describe various kind of energy tariffs are considered top priorities on the EPRI / NIST Smart roadmap for the development of a smart grid [9]. With no common standard in place to build on for TAC Energy, we will use a domain specific language (DSL) that conceptually builds on the work of Tamma et al. [41], an ontology that describes a negotiation process including (i) the involved parties, (ii) the object to negotiate on, and (ii) the negotiation process, i.e. the economic mechanism itself.

Their ontology comprises the following base concepts:

- *Type of Protocol* specifies the mechanism and its specific negotiation rules that will be used for a particular negotiation.
- *Negotiation rule* contains the set of rules that may govern one or more of the aforementioned protocols.

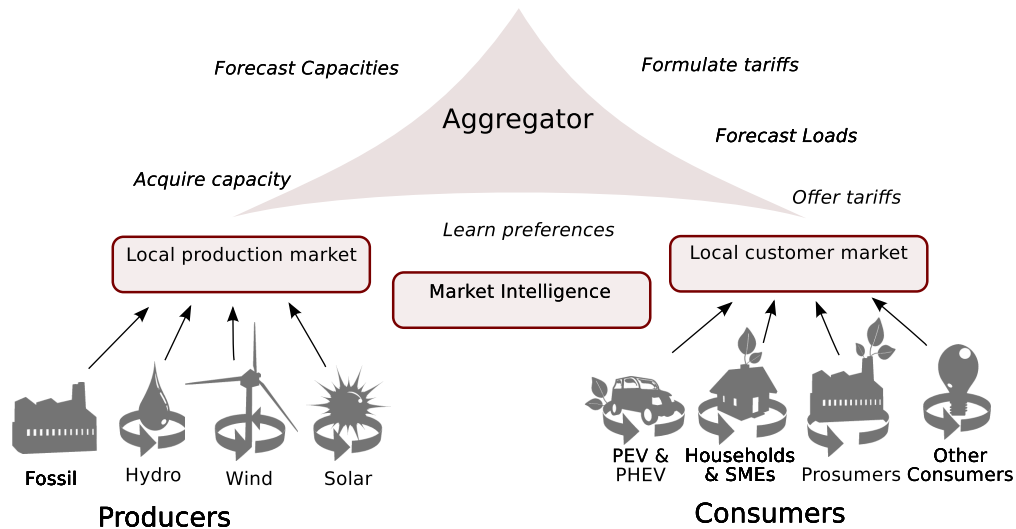


Figure 4: Contracting phase

Protocols contain one or more negotiation rules as parameters.

- *Party* is the concept to describe a single agent that participates in a negotiation along with its role (e.g. buyer, seller). Negotiations can have multiple parties.
- *Negotiation Object* describes the object (in the case of TAC Energy a contract for energy supply or consumption) to be negotiated.
- *Offer* describes a possible combination of negotiation attributes that are expressions of preferences (for example in TAC Energy a consumer tariff could promise a certain energy price for a certain contract runtime, or a specific maximum amount of energy from a controllable generator at a fixed price)
- *Recipe* is a concept from the REA enterprise ontology [15]. It is used to describe the process of reaching agreements on some issues up for negotiation

For TAC Energy we extend this DSL in two particular aspects: First, we add a *Timeslot* concept which is basically a timespan with a defined start and a defined end time, e.g. 00:00 – 23:59 or 14:00 – 14:59. This concept is required as some of the contract attributes are dependent on timeslots, such as energy prices

in a time-of-use consumer tariff. Secondly, the *Negotiation Object Attributes* is added, which contains a standard set of contract (i.e. negotiation object) properties frequently used when designing consumer or generator tariff contracts. This set of properties is not exhaustive and can be extended in future if this becomes necessary, but helps limit complexity for the time being. An example set of contract properties are:

- *Signup-Payment*: a one-time payment (value > 0) or one-time reward (value < 0) paid upon signing a contract. This payment is independent of energy consumption or production.
- *Base-Payment*: a fee (value > 0) or reward (value < 0) paid over time, independent of actual energy consumption.
- *Contract-Duration*: the overall duration of the contract.
- *Leadtime*: the minimum time interval prior to the start of a timeslot s for announcement of price changes that affect s .
- *Timeslot*:
 - *Consumption-Price*: the price for each kWh consumed within the timeslot; contains several additional properties to express e.g. max, min or fixed price.
 - *Generation-Price*: the price (reward) for each kWh fed into the grid within the timeslot; contains several additional properties to express e.g. max, min or fixed price.

This domain specific language is sufficiently flexible and expressive to specify time-of-use tariffs that are offered by aggregators to consumers through a posted price mechanism (see Listing 6), as well as a sealed-bid auction used by a producer to sell his production capacities to an aggregator (see Listing 7). In order to reduce complexity further, the posted-price mechanism is considered the standard for contracting on the consumer side while sealed-bid auctions are considered the standard format for contracting on the production side. Extensions are possible by design, but likely not included in a first version of TAC Energy.

During the contracting phase, an aggregator must develop a portfolio of contracted customers and producers. To do this, aggregators will need to estimate and reason about consumer and generator preferences in order to design appropriate tariffs and to appropriately respond to RFQs respectively. Aggregators will also need to estimate future consumer and generator behavior to build a portfolio that has well-balanced demand and supply over time.

Commonly, companies delegate the tasks of determining customer preferences and estimating business potential for new products (tariffs) to their marketing departments, or they outsource them to specialized service providers. Within the competition scenario, aggregators may request such information from the *market intelligence* service. This service is used somewhat differently for developing tariffs and individual contracts, as described in the next sections.

The market intelligence service also provides aggregators with historic consumption time series for all consumers and generators already under contract. With these time series at hand, an aggregator will be able to estimate how much generation and consumption capacity will be available over time and whether its portfolio is well balanced. Figure 5 shows an example of such a historic time series for a wind turbine with a nominal capacity of 150kW.

3.1.1 Negotiating tariffs

Tariffs are offered contracts that can be accepted or not by anonymous energy consumers and producers. The problem faced by aggregator agents in a competitive market is how to know whether a particular tariff will “sell.” What happens in the real world is that firms are in a continual bidding war with each other, attempting to attract the most “desirable” customers with their offerings. Tariff offers are made in “rounds,” and the number of rounds $|R| > 3$ is indeterminate to prevent “sniping” attacks. In each round $r \in R$, agents are permitted to add or withdraw tariffs from their current offerings, resulting in a set of tariffs U_r for round r . The market intelligence service then runs a customer preference model (under development) to allocate customers to offered tariffs. Each agent is then provided with the number of customers who would agree to

Profile EWF 2003 (Id: 42)

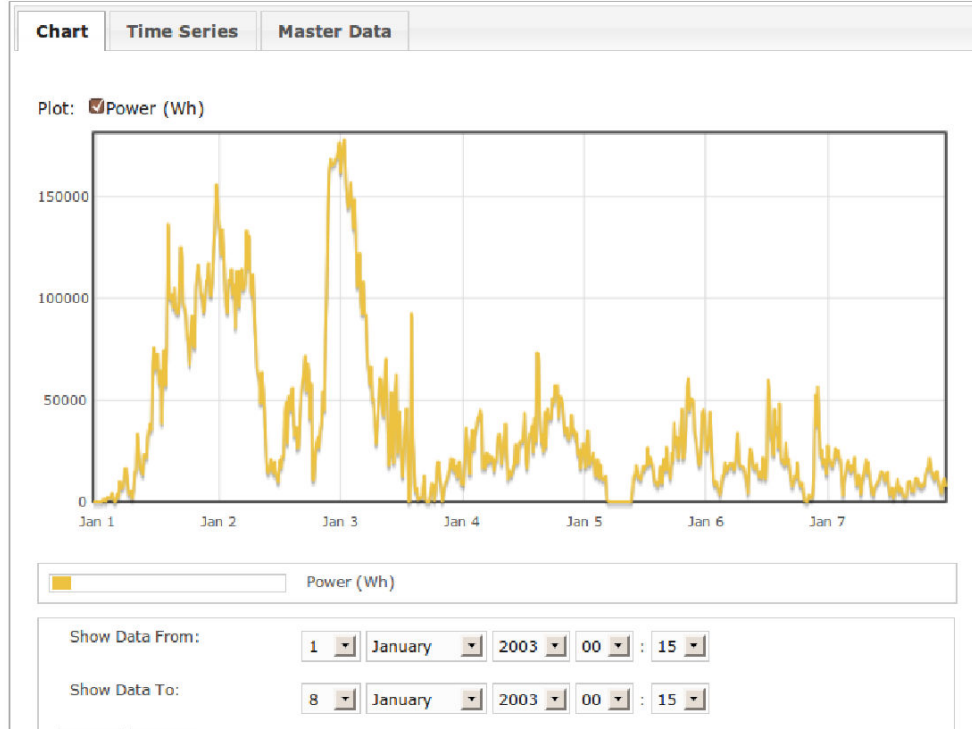


Figure 5: Sample wind turbine generation timeline as provided by the market intelligence service.

each of their offered tariffs, and they may then query the market intelligence service for predicted “demand profiles” for the projected customer base associated with each of their currently offered tariffs. These are simply aggregated time series for the set of customers who currently prefer the individual tariffs. At the end of the last round, no more offerings may be made, and aggregators are charged a fee for each concurrently offered tariff. In other words, in each round r , a set of tariffs $U_{r,a}$ is offered by aggregator a . If the fee for offering a tariff u is m_u^{fee} , then the total tariff fee for the current contracting cycle will be

$$m_{r,a,\text{tariff}}^{\text{fee}} = m_u^{\text{fee}} \max_R(|U_{r,a}|), \forall r \in R. \quad (1)$$

At the end of the last round, all currently-offered tariffs will be available for inspection by all agents through the market intelligence service. Listing 6 is an example of a tariff offering, expressed in an abbreviated version of the contracting language.

```

1 (tacenergy
2   (typeOfProtocol (name "Posted Price")
3     (negotiationRule (name "Sign-Up permitted until") (value 2009-sep-30-23:59))
4   )
5   (party_1 (name "EnergyMart") (role aggregator))
6   (party_2 (name "Julia's wind farm") (role producer) (role consumer))
7   [...]
8   (party_n (name "Ed's SME") (role consumer))
9   (negotiationObject (name "Time-of-use tariff")
10     (negotiationObjectAttributes
11       (contract-duration (2009-oct-1-00:00 2009-dec-31-23:59))
12       (signupPayment 10)
13       (leadtime 00:15)
14       (timeslot (name "day") (06:00 16:59)
15         (consumptionPrice (fixed 0.12))
16         (generationPrice (variable (min 0.07)))
17       )
18       (timeslot (name "evening") (17:00 21:59)
19         (consumptionPrice (variable (max 0.23)))
20         (generationPrice (variable (min 0.09)))
21       )
22       (timeslot (name "night") (22:00 05:59)
23         (consumptionPrice (fixed 0.08))
24         (generationPrice (fixed 0.07))
25       )
26     )
27   (offers
28     (party_2 (offer (type contract-signup)) (timestamp 2009-sep-30-23:50))
29     (party_n (offer (type contract-signup)) (timestamp 2009-sep-30-23:51))
30   )
31   (recipe
32     (party_1 (accept (contract-signup party_2)))
33     (party_1 (accept (contract-signup party_n)))
34   )
35 )

```

Figure 6: Pseudo code for a time-of-use-tariff defined in TAC Energy DSL

3.1.2 Negotiating individual contracts

Individual contracts are negotiated through an RFQ process, initiated by service providers (generators and consumers of power), and proceeding through one or more rounds with as many agents as continue to be interested. The process ends when any party accepts the current contract, or when either the RFQ originator or all aggregators choose to withdraw. The smallest entities that will engage in this process will have capacities of at least 100 times the mean demand of individual households. The specifics of the negotiation

process are undefined at this point, but there are many examples in the literature, such as [18].

Listing 7 is an example of a contract under negotiation, offered by “Ron’s CHP generator” to a set of aggregators including “EnergyMart” and “EnerGreen.”

```

1 (tacenergy
2   (typeOfProtocol (name "Sealed-Bid Auction")
3     (negotiationRule (name "auction-end") (value 2009-sep-30-23:59))
4   )
5   (party_1 (name "Ron's CHP generator") (role producer))
6   (party_2 (name "EnergyMart") (role aggregator))
7   [...]
8   (party_m (name "EnerGreen") (role aggregator))
9   (negotiationObject (name "CHP generation capacity")
10    (negotiationObjectAttributes
11      (contract-duration (2009-oct-1-00:00 2009-dec-31-23:59))
12      (signupPayment -10)
13      (leadtime 00:45)
14      (timeslot (name "day") (00:00 23:59)
15        (generationPrice (variable (min 0.08)))
16      )
17    )
18    (offers
19      (party_2 (offer (type sealed-bid)) (
20        (timeslot (name "day") (00:00 23:59)
21          (generationPrice (variable (min 0.09)))
22        )
23        (timestamp 2009-sep-30-23:50)
24      )
25      (party_2 (offer (type sealed-bid)) (
26        (timeslot (name "day") (00:00 23:59)
27          (generationPrice (variable (min 0.11)))
28        )
29        (timestamp 2009-sep-30-23:50)
30      )
31    )
32    (recipe
33      (party_1 (accept (sealed-bid party_m)))
34    )
35  )

```

Figure 7: Pseudo code for a sealed-bid auction defined in TAC Energy DSL

3.2 Operation phase

During an operation phase, each aggregator must manage the supply and demand resources acquired during the contracting phase, over several consecutive simulated days. Rather than acquiring customers and

generators as during the contracting phase, the aggregator's main task during this timespan is to balance his customer and generator portfolios. The aggregator needs to ensure that the cumulative energy demands and supplies of the customers and generators in his portfolio are balanced at any given point in time throughout the whole operations phase. Deviations between generation and consumption might still occur but will be charged an (expensive) balancing power fee.

In this context, balance between supply and demand means that

$$\forall s \in \mathcal{S}, \sum_{g \in G_a} p_{\gamma,g}(s) = \sum_{c \in C_a} p_{\lambda,c}(s) \quad (2)$$

where $\mathcal{S}$ is the set of discrete timeslots in an operational phase, G_a is the set of contracted power producers for agent a , C_a is the set of contracted consumers for a , $p_{\gamma,g}(s)$ is the total energy produced by generators in G_a during timeslot s , $p_{\lambda,c}(s)$ is the total energy used by consumers in C_a during timeslot s . Note that $p_{\gamma,g}(s)$ can include an arbitrary portion of contracted balancing power as we shall see in Section 3.2.2.

An aggregator's consumer and producer portfolio (i.e. the set of contracts) remains stable throughout an operation phase, but the overall energy demand and supply within the portfolio is volatile over time. The reason is that the actual behavior of generators and consumers is modeled based on historic generation or consumption profiles of corresponding real world entities. A wind turbine under contract with an aggregator will be bootstrapped with a historic time series data of a real wind turbine. Figure 5 shows an example for such a time series. The wind turbine modeled within the competition will expose the same significant generation volatility over time as its real world counterpart did; the same applies to consumers.

3.2.1 Collecting information and predicting the future

As in the contracting phase, an aggregator may request historic time series data for the seven preceeding days (i.e. the length of one complete operation phase) from the market intelligence service for all generators and consumers currently under contract. With this historic data series at hand an aggregator will be able to build its own prediction model for future energy consumption and production of its portfolio, in order to be able to detect and address likely future imbalances.

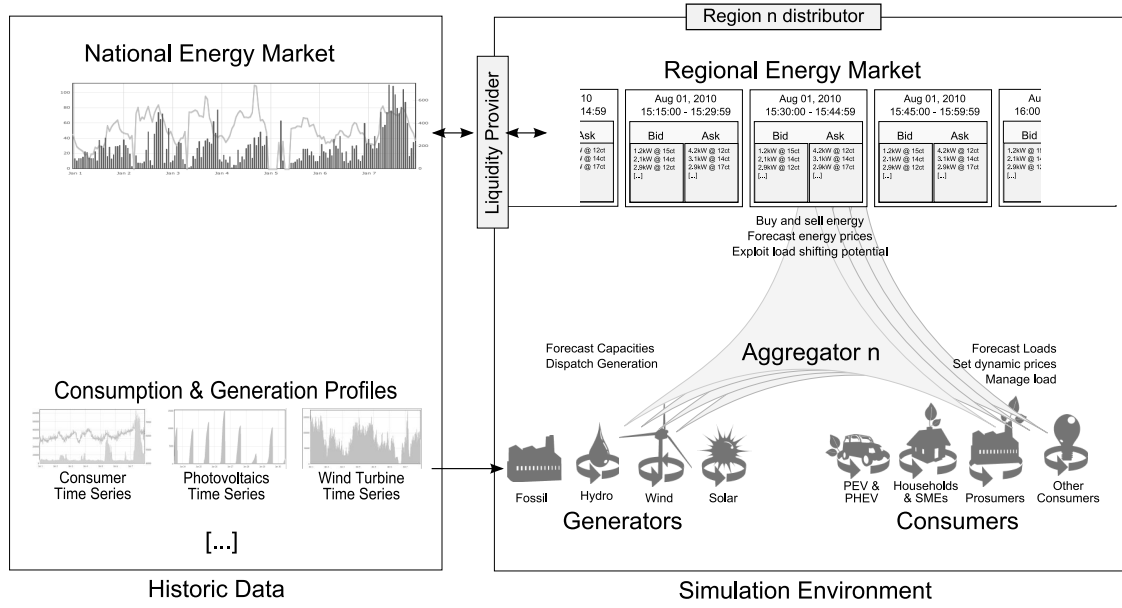


Figure 8: Operational phase

In general, retrieval of future time series data from the market intelligence service is not permitted throughout the competition with one exception: For intermittent generators such as photovoltaics or wind turbines, the estimation of future output solely based on historic time series data is problematic and unrealistic. Predictions about the output of these types of generators are usually based on weather forecasts as described for example in [33].

In order to shield aggregators from having to model weather forecasts, and also because forecasts for specific generators as input for the competition are usually not publicly available³ the approach for this competition is to permit future time series data lookups (only) for *intermittent* generators. Still, in order to model reality closely, these future time series will be artificially distorted (see [1] for background).

Instead of returning the exact future energy production time series for a generator g , an artificial (forecasted) generation $pl_{\gamma,g}(s_j)$ for each timeslot s_j , $s_{n+1} \preceq s_j \preceq s_n$ up to a future timeslot s_n is returned. Each forecasted generation capacity $pl_{\gamma,g}(s_j)$ has a mean error specific to timeslot slot s_j , which is calculated as:

$$pl_{\gamma,g}(s_j) = (1 + \zeta_{\gamma,g}(s_j)) , s_n \prec s_j \preceq s_n \quad (3)$$

³Such forecasts are provided by specialised companies that charge significant fees.

where $\zeta_{\gamma,g}(s_j)$ presents a forecasting error for the generation capacity of generator g in time slot s_j . Forecasting errors increase for forecasts far into the future while they diminish for forecasts close to the present. To model this behavior, $\underline{\phi}$ is assumed to be the mean absolute percentage error (MAPE) of the generation in the first forecasted time slot (s_{n+1}), and $\bar{\phi}$ is the MAPE for the generation in the last forecasted time slot s_η . The expected errors $\phi(s_j)$ for the time slots $s_j, s_{n+1} \prec s_j \prec s_\eta$ are linearly interpolated:

$$\phi(s_j) = \underline{\phi} \cdot \left(1 + \frac{(j-1) \cdot (\bar{\phi} - \underline{\phi})}{\eta - 1} \right) \quad (4)$$

The forecasting error $\zeta_{\gamma,g}(s_j)$ is defined as a stochastic process based on a random walk, in particular a Wiener process [32].

$$\zeta_{\gamma,g}(s_j) = \phi(s_j) + \alpha \sum_{i=n+1}^{j-1} \varepsilon_i \quad (5)$$

Variable α is used to adjust the autocorrelation between two forecasts for generation capacity in adjacent timeslots. Setting $\alpha = 1$ results in a perfect positively autocorrelated time series while $\alpha = 0$ leads to a completely uncorrelated time series. α needs to be adjusted manually for each time series (each intermittent generator) to create forecasts for. The noise variable $\varepsilon_i (n+1 \leq i \leq j)$ is normally distributed $\varepsilon_i \sim N(\mu_i, \sigma_i)$ and the standard deviation σ_i is defined as:

$$\sigma_i = \underline{\phi} \cdot \frac{\sqrt{2\pi}}{2} \cdot \frac{(\eta + i\bar{\phi} - i - \underline{\phi})}{\sqrt{1 + \alpha^2 \cdot i - \alpha^2 \cdot (\eta - 1)}} \quad (6)$$

The idea for the creation of this type of artificial forecasts originally stems from [1] and is described there in detail.

Based on (i) the historic generation schedules of “predictable” generators (e.g. micro turbines or CHP plants), (ii) the historic consumption schedules of the consumers under contract, and (iii) the forecasted generation schedules of intermittent generators, an aggregator will have to predict the estimated future energy generation and consumption schedules for its portfolio as visualized in Figure 9. Note how the uncertainty for the expected demand and supply in time slots far into the future is higher than for those near to the current time slot. In Figure 9(a) the maximum expected overall generation capacity of aggregator a in timeslot s_{n+5} $p_{\gamma}^{max}(a, s_{n+5})$ is much lower than the expected overall load $p_{\lambda}(a, s_{n+5})$. But as the MAPEs for

both numbers are high (indicated as gray boxes), the accuracy of this prediction is low.⁴ 120 Minutes later (Figure 9(b)) the expectation values for supply and demand remained unchanged but the standard deviation decreased. At this point in time the aggregator is able to predict an excess demand situation (e.g. see [23]) for the time slot with a good confidence and thus can now already introduce appropriate countermeasures (e.g. price adjustments or energy trading at the regional market).

3.2.2 Adjusting energy demand and supply

For each timeslot s , each aggregator a must balance expected supply and demand. Demand is the cumulative load, or the sum of the expected loads $p'_{\lambda,c}(s)$ of each consumer c in the the consumer portfolio C_a during timeslot s :

$$p'_{\lambda}(a, s) = \sum_{c \in C_a} p'_{\lambda,c}(s) \quad (7)$$

Supply is total production capacity of all generators g within portfolio G_a , expressed as a minimum without any contracted balancing power, and a maximum including all contracted balancing power.

$$p_{\gamma}^{min}(a, s) = \sum_{g \in G_a} p_{\gamma,g}^{min}(s) \quad (8)$$

$$p_{\gamma}^{max}(a, s) = \sum_{g \in G_a} p_{\gamma,g}^{max}(s) \quad (9)$$

As long as $p_{\gamma}^{min}(a, s) \leq p'_{\lambda}(a, s) \leq p_{\gamma}^{max}(a, s)$, timeslot s is in balance, or at least expected to be so. This is because within the range $[p_{\gamma}^{min}(a, s_n), p_{\gamma}^{max}(a, s_n)]$ production will be adjusted automatically (by dispatching contracted balancing capacity) once timeslot s becomes the current timeslot s_n . The difference between minimum and maximum production $\epsilon(a, s) = p_{\gamma}^{max}(a, s) - p_{\gamma}^{min}(a, s)$ is called the *internal balancing capacity*⁵ of aggregator a for timeslot s . This capacity is simply the sum of the “balancing” portion

⁴Note that the MAPE for the overall consumption stems from the demand forecasting model the aggregator has to build on its own. The MAPE for the overall generation stems in part from the artificial distortion of future generation data as described in this section and in part again from a generation prediction model that the aggregator has to implement for forecasting “non-intermittent” its non intermittent generation capacities like e.g. CHP engines or micro gas turbines

⁵For example a combined heat and power generator, which is technically based on a combustion engine with a nominal output of $p = 50kW$ can adjust its real production within certain boundaries, e.g. $[40kW - 50kW]$.

of production contracts in portfolio G_a (see Section 3.1).

The dispatching of balancing power is done only during the current timeslot s_n . In Figure 9(a), we can see that the forecasted overall consumption of aggregator a deviated from the actual consumption $p_\lambda(a, s_n) = \sum_{c \in C_a} p_{\lambda,c}(s_n)$. But as the difference between $p_\lambda(a, s)$ and $p'_\lambda(a, s)$ was smaller than $\epsilon(a, s)$ the simulation environment automatically dispatched a portion of the internal balancing capacity of the aggregator for this timeslot such that overall demand and supply were balanced, as seen in 9(b). Thus after the timeslot was finished the cumulative demand and supply for this timeslot was balanced. Note that the adjustment is done automatically by the environment. The aggregator agent has no ability to directly affect production or consumption in the current timeslot s_n .

For timeslot s_{n+1} in Figure 9(a), overall demand was forecasted to be within range of the available production capacities even when taking uncertainty (indicated as grey boxes) into account. In other words $p_\gamma^{min}(a, s_{n+1}) \leq p'_\lambda(a, s_{n+1}) \leq p_\gamma^{max}(a, s_{n+1})$. After 2τ simulation time elapsed (Figure 9(b)), the real consumption in this timeslot turned out to be even lower than $p_\gamma^{min}(a, s_{n+1})$. This means that even after the simulation environment reduced the generator's internal balancing power capacity to its minimum level, the overall production still exceeded the overall consumption. In this case *external balancing power* (e.g. some large pumped-storage power plant outside the aggregator's portfolio) had to absorb the excess generated energy. In the energy industry this type of balancing power is usually called an "ancillary service" and its utilization is billed to the aggregator at a defined (high) price. Within the competition this price is defined as a certain percentage above the maximum energy price that occurred at the regional energy exchange for this timeslot.

In slot s_{n+2} in Figure 9(a), a significant difference between overall production and overall consumption is forecast. Internal balancing capacity is likely to be insufficient for leveling the expected difference. In order to avoid the (expensive) utilization of external balancing power, aggregator a can use its contracted pricing power to try to encourage (i) some or all of its consumers $c \in C_a$ to increase their demand, or (ii) some or all of its producers $g \in G_a$ to reduce their production. Technical adjustments initiated by the

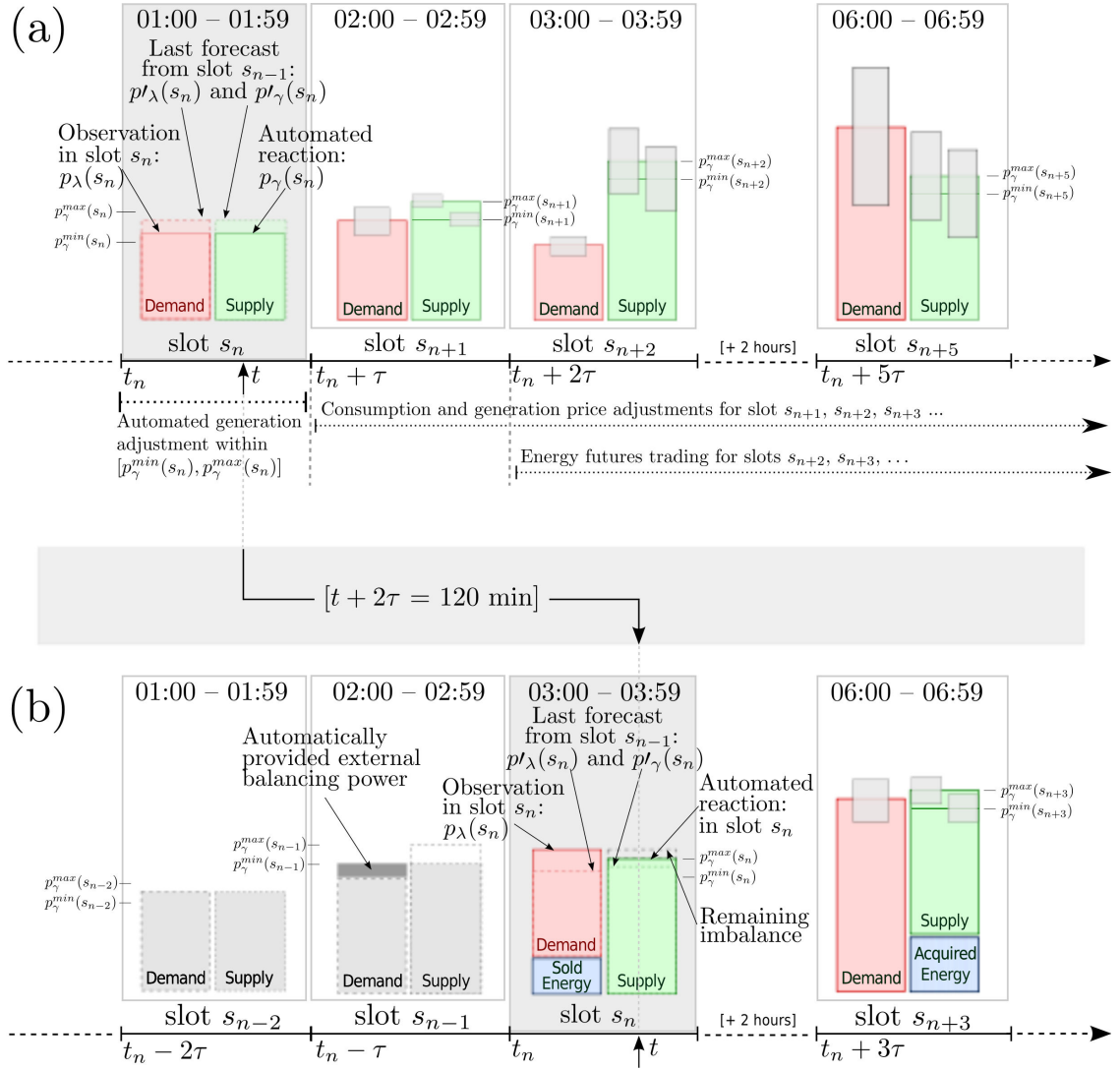


Figure 9: Aggregator's cumulated energy demand and supply as anticipated at a particular point in time.

aggregator (e.g. a remote activation of loads at consumer premises) is not allowed within the competition. In particular, the consumer's forecasted energy consumption c is estimated subject to the energy consumption price for this consumer and this timeslot being $\pi(c, s_{n+2})$

$$\hat{p}_{\lambda,c}(s_{n+2}) = p'_{\lambda,c}(s_{n+2}, \pi(c, s_{n+2})) \quad (10)$$

where $p'_{\lambda,c}(s_{n+2}, \pi(c, s_{n+2}))$ is the predicted load for consumer c in timeslot s_{n+2} , given price $\pi(c, s_{n+2})$.

If the aggregator changes the underlying consumption price to $\pi'(c, s_{n+2})$ the forecasted consumption of

this consumer is expected to increase

$$\hat{p}'_{\lambda,c}(s_{n+2}) = p'_{\lambda,c}(s_{n+2}, \pi'(c, s_{n+2})) \quad (11)$$

The ratio of demand change to price change

$$PE_c = \frac{\hat{p}_{\lambda,c}(\pi) - \hat{p}_{\lambda,c}(\pi')}{\pi - \pi'} \quad (12)$$

is called the “price elasticity” for consumer c . Price elasticities will have to be modeled within the different consumer agents provided by the competition environment following empirical findings on price elasticity as described for example in [38, 36].

Some producers in the aggregator’s portfolio (such as PEV batteries) might have agreed to flexible pricing as well, and therefore their output will be sensitive to price in a similar way. In other words, the minimum generation capacity of aggregator a in timeslot s , $p_{\gamma}^{min}(a, s)$, is likely to decrease if the generation price $\pi_{g,s}$ is changed to $\pi'_{g,s}$ with $\pi'_{g,s} < \pi_{g,s}$. The maximum generation capacity $p_{\gamma}^{max}(a, s)$ will change accordingly such that the internal balancing power capacity $\epsilon(a, s) = p_{\gamma}^{max}(a, s) - p_{\gamma}^{min}(a, s)$ remains unchanged. This is because the balancing portion of production contract are generally not price-sensitive.

3.2.3 Buying or selling futures on the energy market

The re-adjustment of energy prices for consumers and generators as well as the advance reservation of (partial) generator capacity as balancing power reserves are two possibilities to level out an aggregator’s portfolio over time. Besides these two options, a third one is to buy missing or to sell excess capacities the energy market. Within the competition this *market* is modeled as a continuous double auction with uniform pricing and thus resembles the prevalent mechanism design in place for energy spot market trading in Europe and North America [27]. On this market standardized energy futures are traded. An energy future is a binding commitment to consume or to produce a defined amount of energy (e.g. 1kWh) within a defined future time slot (e.g. Aug 01, 2007, 03:00 – 03:59) at a defined price (e.g. 20 ct/kWh). In order to buy or sell energy futures, a market participant sends bid or ask orders to the energy market, which then clears

(matches) all incoming bids and asks at a uniform price determined according to the the maximum turnover principle (see [10] for further details on the mechanism details).

Table 1 shows a sell order before its submission to the energy market (left) and after its clearing (right). In this example Aggregator 1 wants to sell 10 kWh of Power within the timeslot Aug 01, 2007 03:00 – 03:59 at a minimum price of 0.14 EUR/kWh. The offer is set to expire on Aug 01, 2007 01:52h, i.e. if no matching takes place until that time, the order will be canceled. at 01:37 the order is matched on the energy market at a price of 0.15 EUR/kWh and a quantity of 100kWh. Aggregator 1 managed to sell 100kWh of energy for this timeslot, which helps him to better balance his portfolio’s overall energy supply and demand in this timeslot (see Figure 9).

(a) Submitted Sell Order (Ask)		(b) Matched Sell Order	
Order ID:	1	Order ID:	1
Order Owner:	Aggregator 1	Order Owner:	Aggregator 1
Timeslot:	Aug 01, 2007 03:00 – 03:59	Timeslot:	Aug 01, 2007 03:00 – 03:59
Order Type:	Ask	Order Type:	Ask
Order Status:	Open	Order Status:	Matched
Quantity:	100 kWh	Quantity:	100 kWh
Limit:	0.14 EUR / kWh	Limit:	0.14 EUR / kWh
Created at:	Aug 01, 2007 01:37	Created at:	Aug 01, 2007 01:37
Valid Until:	Aug 01, 2007 01:52	Valid Until:	Aug 01, 2007 01:52
Clearing Date:	–	Clearing Date:	Aug 01, 2007 01:42
Clearing Quantity:	–	Clearing Quantity:	100 kWh
Clearing price:	–	Clearing price:	0.15 EUR / kWh

Table 1: Orders submitted to and later matched on the regional energy market.

Besides the aggregators trading energy futures on the regional energy market, there is also a special agent called “liquidity provider.” The liquidity provider is basically representing the point of common coupling (PCC) between simulated region and national grid on the regional market. He can buy energy at the national market and transfer it via the PCC to the simulated region and vice versa. Thus the liquidity provider serves as an arbitrage agent that levels prices of the regional and the national energy market and constitutes an *explicit market coupling* [27].

4 Agent decision problems

Aggregators in the simulation may be individual autonomous agents, as in TAC SCM [7]. They may also be agent communities; for example, it might make sense to separate the contracting and operational behaviors into two separate agents. It is also possible that the “agent,” at least for the contracting phase, could be one or more human decision-makers operating with the assistance of appropriate user interfaces and decision-support tools. This section describes in some detail the decisions such aggregator agents must make to operate within the TAC Energy simulation. We will separate the discussion into the Contracting and Operational phases as described in the previous section. There is very little overlap between the behaviors and decisions during these two phases, except that (i) both phases allow trading in the regional spot market, and (ii) the decisions made during the contracting phase will profoundly influence the required behaviors during the following operational phase.

4.1 Contracting phase

The primary goal of the contracting phase is to acquire access to power sources and customers that result in a portfolio that is profitable and balanced, at least in expectation, over the period of the next operational phase. A secondary goal is to manage financial and supply/demand imbalance risks. For example, an agent will benefit from having reasonably-priced energy sources that can be expected to produce power when demand is expected to be highest within its load portfolio. Predictability is also important, and will generally improve both with volume (because noise as a proportion of demand or supply will be lower with larger numbers of randomly-behaving sources and load, even if they are correlated) and with a balanced portfolio of uncorrelated sources and loads. Risk can be managed by acquiring uncorrelated sources and loads that can be expected to balance each other in real time, by acquiring storage capacity, by acquiring sources that can be used as needed (balancing sources), and by trading futures contracts on the regional exchange.

At the beginning of a contracting phase, an agent will have some number of contracts in force, having

negotiated them earlier. Such contracts have expiration dates beyond the current date. Also, tariffs offered earlier may remain in force; customers who have agreed to a tariff in the past may or may not have an opportunity to opt out and choose a different tariff, and if they have the opportunity they may not choose to exercise it.

4.1.1 Acquire power sources

Power source commitments are obtained by three different methods:

- Large local sources (large wind turbines, wind farms, large CHP plants, etc.) are traded in the local market through the RFQ process as described in Section 3.1.1.
- Small local sources (household and small-business sources) are obtained by offering tariffs in the local market, as described in Section 3.1.2.
- Power from the regional grid is obtained by trading in the regional exchange.

Power sources can be continuous or intermittent, and local continuous sources may have a non-zero balancing component as discussed in Section 3. Continuous sources include power obtained from the regional exchange, as well as the continuous portion of the output from many CHP and hydro plants. Intermittent sources include most renewable sources such as wind and solar plants.

4.1.2 Acquire storage capacity

Storage capacity can be used to absorb excess power or to source power during times of shortage. Power can be absorbed by capacity that is not fully charged, and sourced by capacity that is above its contracted minimum charge level. Storage capacity that is below its minimum charge level is considered to be a load that is possibly responsive to real-time price signals.

Storage capacity can be contracted through the local market through the tariff or the RFQ process. For example, individual owners of PEVs could sign up for tariffs that provide for both charging of the batteries

as well as limited discharging as needed for load balancing by the contracted aggregator. On the other hand, a battery-exchange service for electric vehicles might negotiate a contract for the use of a portion of its current battery inventory for balancing purposes.

4.1.3 Acquire loads

Loads may be contracted through both the local market and the regional exchange, as is the case with power sources.

- Large local loads (industrial facilities and large office parks, for example), could negotiate rates through the RFQ process.
- Small local loads (households and small businesses, for example) must choose tariffs in the local market.
- Agents may choose to sell future power capacity in the regional exchange for periods when it expects to have a surplus. Such advance sales are hard commitments; the sold quantity of power will be transferred out of the system during that interval at a constant rate.

4.2 Operational phase

A detailed timeline of events prior to and during the operation phase is shown in Figure 10. At time t , the simulation is partway through one of the 60-minute timeslots defined by the regional exchange, which started in the past at time t_n . Trading has closed on that slot, and on slot s_{n+1} . Trading will close on slot s_{n+2} at the beginning of slot s_{n+1} at time $t_n + 60$. Between time t and time $t_n + 60$, the agent may continue to trade in slot s_{n+2} and all future slots. The agent may also send price signals to its contracted loads and sources at any time, to the extent allowed in their respective contracts. At the end of each slot (time $t_n + 60$ in this example), the agent will receive information about its supply and demand status at the end of the just-completed slot (slot s_n), and may then send price signals to its contracted sources and loads. These

signals will arrive during the next timeslot (slot s_{n+1} in this example) but will not take effect, regardless of the contract terms, until the beginning of the following timeslot (slot s_{n+2}).

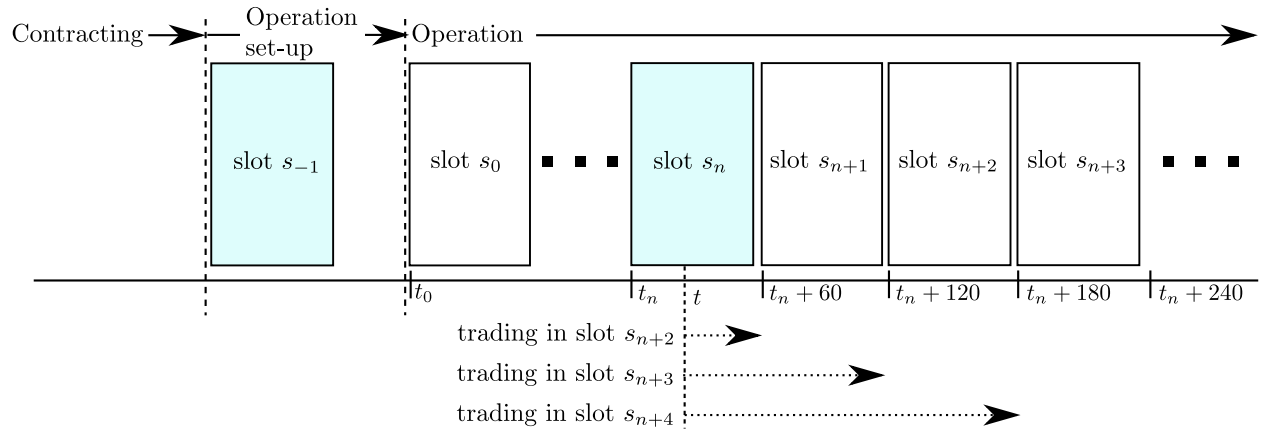


Figure 10: Timeline of interaction between agent and simulation.

4.2.1 Operation set-up

At the end of the contracting phase, the agent has knowledge of its current contract commitments, and of the number of customers who have agreed to its offered tariffs. Operation covers some period of time while these contracts are in place. In order to avoid a lengthy period of adjustment at the beginning of each operation phase, a single timeslot s_{-1} is run in operation mode immediately prior to the period that is to be simulated and evaluated. Results of this preliminary timeslot, along with the current date and time, are then made available to the agent, and the agent is given an opportunity to request history and forecast data, to adjust its variable prices, and to trade in the regional exchange, before full operation commences. During this setup interval, energy can be traded for all timeslots starting with s_0 instead of being restricted to timeslots starting with s_1 , and leadtime restrictions for price changes are waived. This allows the operation phase to begin in a relatively stable state, without requiring agents to spend multiple cycles fine-tuning their balance.

4.2.2 Operation

After the setup period, beginning at time t_0 , the simulation runs continuously. Agents may trade in the exchange and set variable prices at any time. At the end of each timeslot, the agent will receive a performance report giving the supply and demand volume for each of its contracts and tariffs. For each future timeslot $s \succ s_n$ an aggregator a must maintain a forecast of its total expected load and capacity. Total load is the sum of the expected loads $p'_{\lambda,c}(s)$ of each consumer c in the aggregator a 's consumer portfolio C_a for timeslot s . Total production is the sum of expected capacity $p'_{\gamma,g}(s)$ for each producer g within its producer portfolio G_a . Given this information, the agent's task is to adjust prices, and trade in the regional exchange, in order to achieve expected balance.

5 Performance evaluation

In this section we describe in more detail on how to evaluate TAC Energy and its research value, and then we discuss the aggregator profit calculation and winner determination process.

5.1 Research value

With TAC Energy we are building an abstraction of the real world for the particular purposes as outlined before in this paper. Given that this is a proposal for TAC Energy, and not a report on a full implementation, we cannot report laboratory results. Our current plan of work is to finish a specification and prototype in Spring 2010. By the end of Spring we have scheduled a seminar where graduate students will build aggregator agents and compete against each other in an internal course competition. We will use initial results to create a demo for the workshop on "Trading Agent Design and Analysis (TADA) 2010" at the ACM Electronic Commerce Conference at Harvard University. With the input from the TADA audience and other input, we will finalize our specification and implementation and introduce a full competition system in 2011.

After each round of feedback, especially after the first competition in 2011, we will evaluate the TAC

Energy design through the following three questions:

1. How adequate is TAC Energy as a representation of the real world? For instance, we could compare pricing and load balance predictions between TAC Energy and the real world from data available from the MeRegio project.
2. How effectively does TAC Energy support the research agenda of the participating teams? For instance, are teams effective at modeling preferences or price predictions using TAC Energy?
3. How effective is TAC Energy for public policy guidance? Do the suggested solutions provide new insights into real world policy generation?

5.2 TAC Energy winner determination

Within a competition the performance of its participants has to be evaluated and compared at a certain point in time. This is usually accomplished by rank ordering all participants according to one or more defined performance criteria and to declare the best performer in this rank order winner of the competition. This principle also applies to TAC Energy. Consequently this section describes (i) the performance criteria used to rank order the TAC Energy participants, and (ii) the sampling method.

5.2.1 Performance criteria

Each TAC Energy participant (aggregator) is assessed and an overall rank order of all participants is created based on overall profits m^{profit} , calculated as the (monetary) payments, m^{pay} , minus costs, m^{cost} , minus fees, m^{fee} :

$$m^{\text{profit}} = m^{\text{pay}} - m^{\text{cost}} - m^{\text{fee}} \quad (13)$$

- **Payments** are monetary transfers from consumers to aggregators and are based on the agreed contract conditions and the actual (ex-post) measured energy consumptions of the respective consumers as described in Section 3.2.2.

- **Costs** are monetary transfers from aggregators to producers and are based on the agreed contract conditions between the respective producer and aggregator and the actual (ex-post measured) energy produced as described in Section 3.2.2.
- **Fees** are (i) the cost for external balancing power (see Section 3.2.2) used, and (ii) a carbon tax. The carbon tax is a fixed fee (in EUR / kWh) for each kWh of energy produced from non renewable energy sources. The carbon tax remains constant throughout a competition and is publicly announced ahead of the start of the first round.

5.2.2 Sampling method

Several randomly chosen timeslots from each operation phase are selected as “reference timeslots” before a particular competition starts. The chosen timeslots for a particular operation phase are kept secret until that phase ended. Afterwards profits are calculated for each of the reference timeslots of the particular operation phase and then averaged over all reference timeslots. The individual profits of the reference timeslots as well as the average profit of an operations phase are publicly announced immediately after the phase ends. The overall ranking of an aggregator is calculated as the average profit over all reference timeslots from all operation phases. These will be ranked to produce the *winner ranking* at the end of the competition.

6 Conclusion, future work, and research questions

We propose development of a competitive simulation of a market-based management structure for a local energy grid that would closely model reality by bootstrapping the simulation environment with real historic load, generation, weather, and consumer preference and usage data, gathered from over 900 households in the European MEREGIO project [16], and from the American EPRI/NIST Pacific Northwest project [13]. Such a simulation environment would challenge research teams from around the world to write autonomous agents, or agent-assisted decision support systems for human operators [44], that could operate effectively

and profitably in direct competition with each other, while also continuously balancing supply and demand from their portfolios. Teams would also be challenged to exploit the structure of the market, and that structure would be adjusted periodically to defeat counterproductive strategic behaviors. The result would be a body of valuable research data, along with a much higher degree of confidence that such a mechanism could be safely introduced into operating energy systems.

Agents in the proposed market simulation would act as “aggregators,” purchasing power from distributed sources and from regional energy exchanges, and selling power to consumers and exchanges. These agents must solve a set of complex supply-chain problems in which the product is infinitely perishable, and the environment is subject to high variability and uncertainty (e.g. weather effects, equipment and network outages) and limited visibility. They will operate in a dynamic network at multiple timescales, from negotiating long-term contracts with energy sources and tariffs for customers that balance supply and demand in expectation, to day-ahead spot-market trading, to real-time load balancing. They must deal with individual customers and suppliers, while at the same time aggregating the preferences of large groups of customers into market segments for tariff offerings. They must predict supply and demand over monthly, quarterly, and yearly timescales as they develop their portfolios of supplier and customer relationships, and over hours and minutes as they adjust dynamic prices and trade in the spot market in order to maintain real-time balance in the grid.

Competitions have been shown to be an effective way to spur innovation [22, 25, 35]. We expect the primary result of this study to be a clear understanding for policymakers of the capabilities and limitations of open market structures for management of future energy networks that include a variety of distributed sources, including electric vehicles. This simulation will allow such structures to be evaluated in a risk-free environment under a variety of real-world conditions ranging from normal to extreme. The competitive design will effectively uncover potential hazards of proposed market designs in the face of strategic behaviors on the part of the participating agents. The likely effects of various dynamic pricing approaches for consumers and electric vehicle charging can be evaluated. Complex preference models will be developed

that have three important properties: attractive to consumers, attractive to aggregators, and give aggregators sufficient flexibility to balance loads in expectation and in real time. Methods will be developed for charging large numbers of electric vehicles in ways that are both technically feasible and economically attractive to consumers, and that take advantage of the balancing capacities of vehicles.

As a competitive real world testbed, TAC Energy gives ample opportunities for scientists from different disciplines to research and contribute to a variety of important challenges. The initial focus in TAC Energy will be on modeling the behavior and strategies of the aggregator agents. Competing aggregator agents may be fully autonomous, or they may be semi-autonomous, interactive decision support systems [8] for human participants. Preference modeling [6] must be at the core of the market intelligence service since it has to dynamically cluster customer preferences in order to associate agents with tariffs. Currently, lack a good understanding of how to build these multi-attribute utility functions for real world problems, since often the assumptions of utility theory are violated in real world environments. Can we design heuristics that can help us to approximate utility values for rational agents? Another important research question is the representation of the customer preferences through expressive bidding languages. What is the best format? We may need a more expressive bidding/contracting language in the future, such as described by [29, 34]. Once customers are able to express their preferences in a sophisticated way, we should include reputation measures of aggregator agent performance. This should have a direct impact on customer preferences. It may become necessary to learn how to merge different preference models, so that the agents can make overall improved rational decisions.

Since the simulation environment and aggregator agents are subject to high variability, uncertainty and limited visibility, we can study the impact on system stability through exogenous shocks, such as power plant failure, and competition effects among aggregator agents. We can examine the effects of policy changes, such as taxes and incentives. We can research how rapid technical infrastructure changes affect the environment, and how we can balance these changes in real time. Markets are self-organizing mechanisms, and the TAC Energy environment could be an effective tool for research in self-organizing networks and complex

adaptive systems [28].

Our project includes two major phases. In the first phase, we focus our research on the design and development of a competitive simulation for fully autonomous agents, to properly test our market mechanisms and agent strategies. For the second part, we are going to develop a mixed-initiative form of the competition where human players are assisted by intelligent agents (aggregator agents, grid owners, and customers). This competition will help us to research the important topics of preference representation, and elicitation [6], and the tools that are needed for interactive agent assisted decision recommendation.

Notation

A	The set of aggregators
C	Set of consumers
G	Set of generators
t	A particular time, typically the current simulation time (now)
τ	common duration for all timeslots in minutes
S	set of all timeslots
$s \in S$	arbitrary timeslot
s_n	current timeslot
t_n	start time of timeslot s_n
p	energy (typically, power integrated over a timeslot)
p_λ	load (energy consumption)
$p_{\lambda,c}(s)$	load of consumer c in timeslot s
p_γ	production capacity (energy production)
$\epsilon_g(s)$	balancing power capacity for generator g in timeslot s
$\epsilon(a, s)$	overall balancing power capacity of aggregator a in timeslot s
p'_λ	forecasted load (energy consumption)

$p'_{\lambda,c}(s)$	forecasted load of consumer c in timeslot s
$p'_{\lambda}(a, s)$	overall forecasted load of aggregator a in timeslot s
p'_{γ}	foecasted production capacity (energy production)
$p'_{\gamma,g}(s)$	forecasted energy production of generator g in timeslot s
$p'_{\gamma}(a, s)$	overall forecasted energy production of aggregator a in timeslot s
$\zeta_{\gamma,g}(s)$	Forecasting error for energy production of generator g in timeslot s
π, π'	prices
$\pi_{g,s}$	price for production, generator g , timeslot s
$\pi_{c,s}$	price for consumption, consumer c , timeslot s
$\hat{p}_{\gamma,g}(\pi, s)$	Forecast for generator g in timeslot s , given production price π
PE_g	Price elasticity for generator g
$\hat{p}_{\lambda,c}(\pi, s)$	Forecast for consumer c in timeslot s , given consumption price π
PE_c	Price elasticity for consumer c
m	money
m^{profit}	aggregator profit
m^{pay}	received payments – costs from the standpoint of customers
m^{cost}	costs
m^{fee}	fee – negative when customers pay them
r	a round in the tariff-publishing process
U	a set of tariffs
m_u^{fee}	The fee for publishing tariff u

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